

Department of Computer Science and Engineering
Academic Year 2021– 2022(Odd Semester)

Degree, Semester & Branch: B.E, V & CSE

Course Code & Title: CS8501 & Theory of Computation

Name of the Faculty member (s): Dr. I Gethzi Ahila Poornima

Innovative Practice Description

- **Unit / Topic:** Unit III / PushDown Automata
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO 3
- **Activity Chosen:** Active Learning- Cross Word Puzzle
- **Justification:**

Pushdown automata is one of the important concepts in Theory of Computation. Since it plays a very important role in designing the parsing phase of the compiler, the awareness about the PDA is necessary.

- **Time Allotted for the Activity:** 20 Minutes
- **About the Event**

A crossword is an interesting word puzzle activity which makes the students to recollect the concept and try to find out the answer. The topic for this event should be informed to the students well in advance. Questions for the chosen topic will be given as cross word puzzle in the class. Time will be allotted based on the number of questions and level of questions. Subsequently, the correct answers have to be discussed in the same class.

- **Details of the Implementation:**

The activity was conducted in the class room with 52 number of students. The announcement was given to the students in-prior. In the class room, the printed sheet of crossword puzzle of 10 questions to each student separately to the students. The students filled the sheets and returned. The answers for the wrong questions were discussed in the later session of the same class immediately to make them to know the answer for all the questions and clear in this particular topic. Figure 1 shows sample cross word puzzle sheet and picture taken during the event. Around 80 percentage of students answered all the questions correctly.

- **CO – PO / PSO mapping:**

CO	PO2	PO3	PSO1
CO5	1	2	2

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO2	PO3	PSO1
	1	2	2
Justification for correlation	Students will be able to formulate a model for a system and interpret by understanding the context free grammar and design of push down automata.	Students could verify and validate the functionality of a system using the knowledge of Context free grammar and push down automata.	Students will be able to design software components such as syntax analyzer of compiler using the context free grammar and push down automata.

• **Images / Screenshot of the practice:**

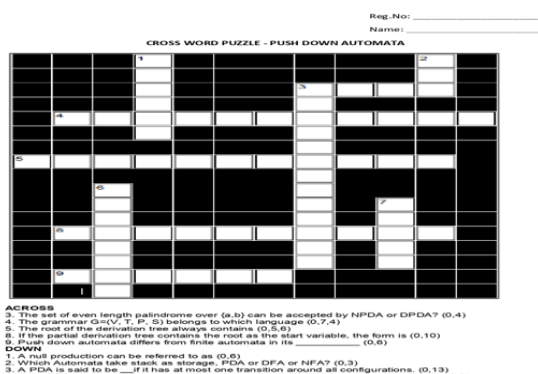


Figure 1: Sample sheet & picture taken

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- Most of the students felt that this event helped them to recollect the nature of PDA and able to compare with FA.

❖ **Benefit of the practice:**

- All the students enthusiastically answered the puzzle activity.
- They can easily remember the basic terminologies and keywords used in PDA and its characteristics.

❖ **Challenges faced in implementation:**

- Few students do not recollect the PDA and its nature.

Signature of Faculty Member

HOD